# The curvature trichotomy of medial-axis reconstruction: ideal versus finite centres and the sign of the Busemann Hessian*

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**Abstract**

For a closed set $X$ in a complete Riemannian manifold the medial-axis *reconstruction* $\mathcal{R}_X = \bigcup_{a \in M_X} B(a, r(a))$ is the union of the open medial balls. Białożyt's Theorem 6 [1] characterises $\mathcal{R}_X$ for $X \subseteq \mathbb{R}^n$ as the convex hull, less $X$, together with the interiors of the *defective* supporting half-spaces — those whose contact set is non-convex. Across the companion notes this characterisation took two seemingly different faithful forms: in non-positive curvature an exact, *strict* description by horoballs with a Chebyshev/horoconvex defect [2, 3, 4], and in positive curvature its total collapse, $\mathcal{R}_X = M \setminus X$ [5]. We show these are the two branches of a single *trichotomy*, governed by one invariant. The maximal free convex region carrying the reconstruction near a point — the horoball analogue — is a sublevel set of a Busemann function, and the sign of its Hessian, fixed by the sign of sec through the Hessian comparison theorem, determines whether its centre is *ideal* or *finite*, and with it the entire detection problem:

- $\sec \leq 0$: $\operatorname{Hess} b \succeq 0$, Busemann functions are convex, horoballs are convex and *unbounded* — an *ideal* centre at $\xi \in \partial_\infty M$. Detection is a convexity question: $\mathcal{R}_X$ is the hull plus the defective horoballs, where "defective" means the contact set differs from its horoconvex hull (Chebyshev on flat horospheres, Carnot on homogeneous ones), and $\mathcal{R}_X \subsetneq M \setminus X$ strictly.
- $\sec \equiv 0$: $\operatorname{Hess} b \equiv 0$, $b$ is affine, the maximal free convex region is a *half-space* with its ideal centre at infinity — the knife-edge, Motzkin detection, Białożyt's theorem verbatim.
- $\sec > 0$: $\operatorname{Hess} b \preceq 0$, $b$ is concave; convex free regions cannot grow unbounded (Bonnet–Myers; point soul), so the maximal one is a bounded ball with a *finite* centre, equidistant from its boundary. Detection collapses to $\geq 2$ contacts — and on the round sphere every point is captured: $\mathcal{R}_X = M \setminus X$.

The flat case sits exactly on the boundary between the two: it is the $R \to \infty$ limit in which the finite circumcentre recedes to the ideal point and the convexity radius diverges. The trichotomy thus explains, by the single mechanism "can a free convex region be unbounded," why reconstruction is partial and convexity-detected below curvature 0, total and contact-counted above it, and Białożyt's theorem the boundary between them.


*Capstone of a series generalising Theorem 6 of [1]: [2] carried its sufficiency half to Hadamard manifolds, [3] proved the full characterisation in real hyperbolic space, [4] gave the faithful curvature-free characterisation for every Hadamard manifold and isolated the Gauss obstruction, and [5] crossed into positive curvature. This note unifies them: it identifies the single invariant — the sign of the Busemann Hessian, i.e. the sign of the sectional curvature — that governs the whole theory, and reads each companion result as one branch of the resulting trichotomy. Drafted with the assistance of Claude (Anthropic); all arguments were re-derived and checked by hand, and the positive-curvature branch verified numerically (convexity-radius threshold and Hessian sign-flip on $\mathbb{S}^2$; zero failures). Remaining errors are ours.

# Contents



# 1 Introduction

## One theorem, two faces

For a closed set $X$ in a complete Riemannian manifold $M$ write $M_X$ for its *medial axis* — the points with two or more nearest points of $X$ — and, for $a \in M_X$, $r(a) = \operatorname{dist}(a, X)$ for its reach. The *reconstruction* of $X$ is the union of the open medial balls,

$$\mathcal{R}_X = \bigcup_{a \in M_X} B\big(a, r(a)\big) \;\subseteq\; M \setminus X.$$

Białożyt's Theorem 6 [1] characterises $\mathcal{R}_X$ for closed $X \subseteq \mathbb{R}^n$: with $\widehat{X} = \overline{\operatorname{conv}}\, X$,

$$\mathcal{R}_X = (\widehat{X} \setminus X) \;\cup\; \bigcup \big\{ \operatorname{int} H : \; H \text{ a supporting half-space, } X \cap \partial H \neq \operatorname{conv}(X \cap \partial H) \big\}. \tag{1}$$

The companion notes carried this statement across the curvature spectrum, and on its face the two ends look unrelated.

*Below zero.* In every Hadamard manifold ($\sec \le 0$) the half-space becomes a horoball, conv becomes a horoconvex hull hconv built from the Busemann geometry of $M$, and (1) holds verbatim:

$$\mathcal{R}_X = (\widehat{X} \setminus X) \;\cup\; \bigcup \big\{ \operatorname{int} \mathcal{H} : \; X \cap \partial \mathcal{H} \neq \operatorname{hconv}(X \cap \partial \mathcal{H}) \big\} \tag{2}$$

[4, Thm. 5.6–5.7], with hconv = conv on flat horospheres (so (2) contains (1) and the hyperbolic theorem [3]) and the Carnot hull on homogeneous ones. The reconstruction is a *strict* subset of $M \setminus X$, and the unreconstructed points are exactly those whose nearest-point ray escapes to infinity carrying a single foot.

*Above zero.* On the round sphere ($\sec \equiv +1$) the same engine gives the opposite conclusion: for every closed $X \subseteq \mathbb{S}^n$ with $|X| \ge 2$,

$$\mathcal{R}_X = \mathbb{S}^n \setminus X \tag{3}$$

[5, Thm. 4.1] — reconstruction is *total*, the certificate (2) strictly undercounts, and the convexity defect that organises (1)–(2) has no analogue: a convex spherical cap reconstructs its entire complement [5, Prop. 5.1].

## The invariant that unifies them

The purpose of this note is to show that (2) and (3) are not two theorems but two *branches* of one, selected by a single invariant. Both proofs run the same machine. Outside the hull, the reconstruction of a point $p$ is carried by a *maximal free convex region* $\mathcal{H}$ — the largest geodesically convex open set that avoids $X$ and contains $p$, the role played by a horoball in $\mathbb{H}^n$ and by a free metric ball on $\mathbb{S}^n$. The region $\mathcal{H}$ is a sublevel set $\{b \le c\}$ of a *Busemann function* $b$ (the signed distance to its bounding "horosphere"), and *everything* about the theory — where the centre of $\mathcal{H}$ sits, what "defective" means, whether $\mathcal{R}_X$ is partial or total — is read off the geometry of $\mathcal{H}$. That geometry is fixed by one number: the sign of $\operatorname{Hess} b$.

> **Trichotomy (Theorem 3.1).** The sign of $\operatorname{Hess} b$ equals the sign of sec (Hessian comparison), and it controls the *centre type* of the maximal free convex region, hence the detection mechanism:

| | $\operatorname{Hess} b$ | region $\mathcal{H}$ | centre / detection |
|---|---|---|---|
| $\sec \le 0$ | $\succeq 0$ (convex $b$) | convex, unbounded | ideal $\xi \in \partial_\infty M$ / Chebyshev hull |
| $\sec \equiv 0$ | $\equiv 0$ (affine $b$) | half-space | $\infty$ / Motzkin convexity |
| $\sec > 0$ | $\preceq 0$ (concave $b$) | bounded ball | finite $a \in M$ / $\ge 2$ contacts |

The mechanism is a single dichotomy — *can a free convex region be unbounded?* — and the answer is dictated by curvature. In $\sec \le 0$, convexity of $b$ lets horoballs run to infinity, so their centre is an ideal point $\xi$; a finite reconstructing centre must *drift* toward $\xi$, and it is medial precisely when its foot is non-Chebyshev in the contact set — a genuine convexity test (Motzkin on flat horospheres, Carnot on curved ones), which is what makes $\mathcal{R}_X$ partial and the defect the organising notion. In $\sec > 0$, concavity of $b$ forbids unbounded convex sublevel sets (Bonnet–Myers compactness, or a point soul): the maximal free convex region is a *bounded* ball, its centre a genuine *finite* point equidistant from the boundary, hence medial as soon as the boundary carries two contacts — no convexity test at all, the detection problem trivialised. The flat case $\sec \equiv 0$ is exactly the boundary: $b$ is affine, the "ball" is a half-space, the finite circumcentre has receded to the ideal point at infinity and the convexity radius has diverged — the unique knife-edge on which Białożyt's theorem lives. Pushing past it into $\sec > 0$, the recovered finite centre swallows so much that reconstruction becomes total.

This is the sense in which Theorem 6 has not one generalisation but a *trichotomy*: the same hull-plus-defective-region template, with the region's centre ideal or finite according to the sign of the Busemann Hessian, and the detection collapsing from a convexity question to a counting one as curvature crosses 0.

## Organisation

Section 2 fixes the common objects — medial axis, Busemann functions, maximal free convex regions, the defect template. Section 3 states and proves the trichotomy theorem, the conceptual core: the Hessian-comparison input, the three centre-type conclusions, and the finite-equidistant-centre principle that trivialises detection above zero. Sections 4–6 then read each branch as an instance, quoting the companion notes for the detailed reconstruction theorems and giving the mechanism in trichotomy language: the Hadamard branch (§4, ideal centres, the faithful hconv characterisation and its Carnot/Alexandrov frontier), the flat knife-edge (§5, Białożyt as the degenerate limit of both branches), and the positive branch (§6, finite centres, total reconstruction, the contact-count collapse, and what survives off constant curvature). Section 7 collects the unified picture, the phase transition at $\sec = 0$, and the open problems each branch bequeaths.

## 2 The common objects

Throughout $M$ is a complete Riemannian manifold, $X \subseteq M$ is closed and nonempty, $d$ is the geodesic distance, and $B(a,\rho) = \{x : d(x,a) < \rho\}$. We recall $M_X$, $r(a) = d(a,X)$ and $\mathcal{R}_X = \bigcup_{a\in M_X} B(a,r(a))$ from §1; the medial balls are free ($B(a,r(a))\cap X = \varnothing$), so always $\mathcal{R}_X \subseteq M\setminus X$. Two structural objects underlie all three branches.

**Busemann functions and the horoball analogue.** A *Busemann function* is a normalised limit of distance functions. In $\sec \le 0$ one fixes a geodesic ray $\gamma$ to an ideal point $\xi \in \partial_\infty M$ and sets $b_\xi(x) = \lim_{t\to\infty}\big(d(x,\gamma(t)) - t\big)$; the limit exists, $b_\xi$ is 1-Lipschitz, and its sublevel sets $\mathcal{H} = \{b_\xi \le c\}$ are the *horoballs* at $\xi$. In $\sec > 0$ there are no rays — the manifold is compact (Bonnet–Myers, if $\sec \ge \kappa > 0$) — and the role of $b_\xi$ is played by a *signed distance to a sphere*: on $\mathbb{S}^n$ the function $b_v(x) = d(x,v) - \frac{\pi}{2}$, the signed distance to the equator $\partial H_v = S(v,\frac{\pi}{2})$ of a pole $v$, is the exact analogue — its zero set is the "horosphere," its negative sublevel set $\{b_v < 0\} = B(v,\frac{\pi}{2})$ the free hemisphere, and $v$ the (now finite) centre [5, §2,§9]. In all cases we call $\mathcal{H} = \{b \le c\}$ a *maximal free convex region* when it is geodesically convex, disjoint from $X$, and maximal with these properties among the sublevel sets of its Busemann function; its boundary $\partial\mathcal{H}$ is the *horosphere*, its *contact set* is $K = X \cap \partial\mathcal{H}$, and its *centre* is the ideal point $\xi$ (when $\mathcal{H}$ is unbounded) or the finite circumcentre $a$ with $\mathcal{H} = B(a,r(a))$ (when bounded).

**The defect template.** Every branch instantiates the same shape of theorem,

$$\mathcal{R}_X = (\widehat{X}\setminus X) \;\cup\; \bigcup\big\{\operatorname{int}\mathcal{H} :\; \mathcal{H} \text{ maximal free, } \textit{defective}\big\}, \tag{4}$$

where $\widehat{X}$ is the hull (the complement of the union of all open free convex regions) and "defective" is a condition on the contact set $K$ alone, decidable from $X$ and the geometry of $M$ with $M_X$ absent. The content of the trichotomy is what "defective" *is*, and whether (4) is a strict description or collapses to $M\setminus X$. We will see that this is decided entirely by the centre type of $\mathcal{H}$.

**The reconstruction engine, curvature-free.** Two inputs are common to all branches and use no curvature sign beyond what is named.

**Theorem 2.1** (Hull inclusion and the swallow; [2, Thm. 3.1], [4, Lem. 5.5],[5, Lem. 3.1])**.** *For every complete $M$ and closed $X$:*

1. (Hull.) $\widehat{X}\setminus X \subseteq \mathcal{R}_X$.
2. (Swallow.) *If a maximal free convex region $\mathcal{H}$ admits medial centres $a_k \in M_X$ converging to its centre (drifting to $\xi$ in the unbounded case, or a single finite medial centre in the bounded case), then* $\operatorname{int}\mathcal{H} \subseteq \mathcal{R}_X$*. The estimate is $d(p,a_k) - r(a_k) \to b(p) < 0$ for $p \in \operatorname{int}\mathcal{H}$ and uses only horofunction convergence* — no curvature bound.

What the engine leaves open is exactly the certificate: *which* $\mathcal{H}$ admit such medial centres, read from $X$. That is the detection problem, and the next section shows it is governed by the sign of $\operatorname{Hess} b$.

## 3 The trichotomy theorem

We isolate the single comparison-geometric fact behind the whole theory. Recall the Hessian comparison theorem (e.g. [7, Ch. 6]): for $r(\cdot) = d(\cdot, o)$ and the model space form of curvature $\kappa$,

$$\sec \le \kappa \implies \operatorname{Hess} r \succeq \operatorname{Hess} r_\kappa, \qquad \sec \ge \kappa \implies \operatorname{Hess} r \preceq \operatorname{Hess} r_\kappa,$$

on the tangential subspace $(\nabla r)^\perp$, with the model $\operatorname{Hess} r_\kappa = \mathfrak{c}_\kappa(r)\,(g - dr \otimes dr)$ and $\mathfrak{c}_\kappa = \sqrt{\kappa}\cot(\sqrt{\kappa}\,r)$ for $\kappa > 0$, $= 1/r$ for $\kappa = 0$, $= \sqrt{-\kappa}\coth(\sqrt{-\kappa}\,r)$ for $\kappa < 0$. Passing to the Busemann limit $r \to \infty$ at $\kappa = 0$ sends the model term $1/r \to 0$, leaving the sign of $\operatorname{Hess} b$ equal to the sign forced on $\operatorname{Hess} r$ by the curvature sign.

**Theorem 3.1** (Curvature controls the centre)**.** *Let $M$ be complete, $X \subseteq M$ closed, and $\mathcal{H} = \{b \le c\}$ a maximal free convex region with Busemann function $b$. The sign of* $\sec$ *fixes the sign of* $\operatorname{Hess} b$ *and, through it, the centre type of $\mathcal{H}$.*

(i) (sec $\le 0$: ideal centre.) $\operatorname{Hess} b \succeq 0$*: $b$ is convex, $\mathcal{H} = \{b \le c\}$ is a convex sublevel set containing the ray to $\xi$, hence* unbounded*, with the ideal centre $\xi \in \partial_\infty M$. A finite centre is never equidistant from $\partial\mathcal{H}$; it is medial only after* drifting *toward $\xi$, and then iff its foot is non-Chebyshev in $K$.*

(ii) (sec $\equiv 0$: ideal centre at infinity.) $\operatorname{Hess} b \equiv 0$*: $b(x) = -\langle x, u\rangle + \mathrm{const}$ is affine, $\mathcal{H}$ is a half-space, the unique convex maximal free region, with centre the ideal point $u$ at infinity.*

(iii) (sec $> 0$: finite centre.) $\operatorname{Hess} b \preceq 0$*: $b$ is concave, so its convex sublevel sets cannot grow unbounded. Quantitatively, every geodesically convex set has diameter at most the convexity radius $\le \pi/(2\sqrt{\sup\sec})$ (finite, e.g. $\frac{\pi}{2}$ on $\mathbb{S}^n$); if* $\sec \ge \kappa > 0$ *then $M$ itself has diameter $\le \pi/\sqrt{\kappa}$ (Bonnet–Myers), and if $M$ is open with* $\sec > 0$ *it has a* point *soul (Cheeger–Gromoll [8]; Perelman [9]), so it has no unbounded totally convex proper subset. Hence $\mathcal{H}$ is a* bounded *ball $B(a, r(a))$ with a* finite *centre $a$, equidistant (at $r(a)$) from $\partial\mathcal{H}$.*

*Proof.* The sign of $\operatorname{Hess} b$ is the Hessian comparison theorem in the Busemann limit, as recalled above. For (i): on a Hadamard manifold Busemann functions are convex [6, II.8.24], so $\{b \le c\}$ is convex; it contains the asymptotic ray $\gamma$ (along which $b$ decreases linearly), hence is unbounded, and $\xi = \gamma(\infty)$ is its ideal centre. A finite point $a$ has $b$ strictly convex transversally, so $\partial\mathcal{H} = \{b = c\}$ is at non-constant distance from $a$ — $a$ is not a circumcentre — which is exactly why detection needs the contact set to fail Chebyshev: see §4. (ii) is the flat specialisation, $\operatorname{Hess} b \equiv 0$ forcing $b$ affine, $\mathcal{H}$ a half-space. For (iii): the sign $\operatorname{Hess} b \preceq 0$ is comparison with $\kappa = \sup\sec > 0$, under which $\mathfrak{c}_\kappa(r) = \sqrt{\kappa}\cot(\sqrt{\kappa}\,r)$ changes sign at the convexity radius $r = \pi/(2\sqrt{\kappa})$; beyond it $b$ is concave and $\{b \le c\}$ ceases to be convex, so a *convex* free region cannot extend past a ball of that radius. The named global statements — convexity radius finite, Bonnet–Myers, point soul — each independently force every convex (indeed totally convex) free region bounded; a bounded convex region is inscribed in a smallest enclosing metric ball $B(a, r(a))$, whose finite centre $a$ is equidistant from the bounding sphere by construction. □

The operational half of the trichotomy is the contrast in *detection* that the centre type forces. We state the two principles; each is proved in its branch.

**Proposition 3.2** (Finite equidistant centre trivialises detection — any manifold)**.** *A free geodesic ball $B(a, \rho)$ with $\rho = d(a, X)$ has* every *boundary point at distance $\rho$ from $a$. Hence if the contact* *set $K = X \cap \partial B(a,\rho)$ has two or more points, then $a \in M_X$ and $B(a,\rho) \subseteq \mathcal{R}_X$ — with no condition on the position or geometry of the contacts. A finite equidistant centre never asks "which contact is nearest," the question whose answer is the Chebyshev structure, so that structure plays no role.*

*Proof.* For $x \in \partial B(a,\rho)$, $d(a,x) = \rho = d(a,X)$, so every contact is a foot; two contacts give two feet, $a \in M_X$ with $r(a) = \rho$, and every point of $B(a,\rho)$ lies in this medial ball [5, Prop. 3.3]. □

**Proposition 3.3** (Ideal centre demands a convexity test — $\sec \le 0$)**.** *Let $\mathcal{H}$ be a maximal free horoball at $\xi$ with contact set $K$ on its horosphere. A finite centre high over a horosphere point $\sigma$ is medial iff $\sigma$ has two or more nearest contacts in the intrinsic geometry of $\partial\mathcal{H}$; and a drift to $\xi$ reconstructing $\operatorname{int}\mathcal{H}$ exists iff $K \neq \operatorname{hconv}(K)$. Detection is the horoconvex-hull (Chebyshev/Motzkin) question.*

*Proof.* This is the deepest-shadow necessity and the swallow sufficiency of [4, Thm. 5.6]; see §4. The point is that, $\mathcal{H}$ being unbounded, the centre is *not* equidistant from $\partial\mathcal{H}$ (Theorem 3.1(i)), so being medial is a genuine nearest-among-many condition — non-Chebyshev — rather than the trivial count of Proposition 3.2. □

*Remark* 3.4 (The Hessian sign-flip, concretely and numerically). On $\mathbb{S}^n$ the contrast is visible in one formula. For $b_v(x) = d(x,v) - \frac{\pi}{2}$,

$$\operatorname{Hess} b_v = \cot\big(d(x,v)\big)\,\big(g - dr \otimes dr\big),$$

positive inside the $v$-hemisphere ($d < \frac{\pi}{2}$, where the would-be horoball is convex), vanishing on the equator ($d = \frac{\pi}{2}$), and *negative* outside ($d > \frac{\pi}{2}$): the signed-distance "Busemann" function is concave past the convexity radius, so the maximal free convex region is exactly the hemisphere $B(v, \frac{\pi}{2})$ — bounded, finite centre $v$. Both facts were checked numerically on $\mathbb{S}^2$: a cap of angular radius $R$ is geodesically convex iff $R \le \frac{\pi}{2}$ (convex at $R = 1.551$, leaking at $R = 1.591$), and the tangential eigenvalue of $\operatorname{Hess} d(\cdot, v)$ matches $\cot r$ to four places, flipping sign exactly at $r = \frac{\pi}{2}$. This is the $\kappa = +1$ instance of Theorem 3.1(iii).

*Remark* 3.5 (Why the flat case is a knife-edge). The two branches meet only at $\sec \equiv 0$, and degenerately. View $\mathbb{S}^n_R$ of radius $R$ (curvature $R^{-2} > 0$): the convexity radius is $\frac{\pi}{2}R \to \infty$ and the focal circumcentre of a free hemisphere recedes to distance $\frac{\pi}{2}R \to \infty$ as $R \to \infty$ [5, Rem. 3.2], so the finite centre escapes to infinity and the bounded ball opens into a half-space — exactly the $\sec = 0$ picture of (ii). Symmetrically, sharpening $\sec \le -a^2 < 0$ contracts horoballs toward $\xi$; relaxing $a \to 0$ opens them to half-spaces. The flat case is the unique fixed point of both limits: $\operatorname{Hess} b \equiv 0$, neither convex nor concave, the ideal centre sitting at the finite/infinite boundary. It is no accident that Białożyt's theorem — the one case with a clean conv certificate that is simultaneously strict and Motzkin-detected — lives precisely there.

# 4 The branch $\sec \le 0$: ideal centres and the Chebyshev defect

In non-positive curvature Theorem 3.1(i) gives unbounded horoballs with ideal centres, and Proposition 3.3 makes detection a convexity question. The faithful characterisation is the template (2), proved in [4] and quoted here in trichotomy form.

**Definition 4.1** (Horospherical convex hull; [4, Def. 4.1]). *For a maximal free horoball $\mathcal{H}$ at $\xi$ with horosphere $S = \partial\mathcal{H}$ and an ideal direction $u$, let $L_u(y) = \frac{d}{d\varepsilon}\big|_{0^+} b_{\xi_\varepsilon}(y)$ be the one-sided derivative of the Busemann function along a tilt $\xi_\varepsilon \to \xi$. The* horospherical convex hull *of $K \subseteq S$ is*

$$\operatorname{hconv}(K) = \big\{\sigma \in S : \ L_u(\sigma) \geq \inf_K L_u \ \textit{for every } u\big\},$$

*the trace on $S$ of the intersection of the tilted horoballs containing $K$. It is a genuine hull, built from the Busemann geometry of $M$ alone, equal to* conv *on a flat horosphere and to the Carnot hull on a homogeneous one.*

**Theorem 4.2** (Faithful characterisation in sec $\leq 0$; [4, Thm. 5.6–5.7]). *For every Hadamard manifold $M$ and every closed $X \subseteq M$,*

$$\mathcal{R}_X = (\widehat{X} \setminus X) \ \cup \ \bigcup\big\{\operatorname{int}\mathcal{H} : \ X \cap \partial\mathcal{H} \neq \operatorname{hconv}(X \cap \partial\mathcal{H})\big\},$$

*the defect certificate — the contact set is not horospherically convex — being decidable from $X$ and the geometry of $M$, with $M_X$ absent. It reduces to $X \cap \partial\mathcal{H} \neq \operatorname{conv}(\cdot)$ on flat horospheres ($\mathbb{R}^n$, $\mathbb{H}^n$) and to its Carnot form on homogeneous ones. The reconstruction is strict: $\mathcal{R}_X \subsetneq M \setminus X$ whenever $X$ has a point whose nearest-point ray escapes to infinity with a single foot.*

The two analytic inputs are both curvature-free and both visible in the trichotomy. *Necessity* is Danskin's first-order condition for the depth $D(\xi) = \inf_x b_\xi(x) - b_\xi(p)$: the deepest free horoball over $p$ has its foot $\sigma(p)$ in $\operatorname{hconv}(K) \setminus K$, so $K \neq \operatorname{hconv}(K)$ [4, Thm. 5.6]. *Sufficiency* is the swallow (Theorem 2.1(2)) along the medial centres drifting to $\xi$ — which exist precisely because, the centre being ideal (Theorem 3.1(i)), a non-horoconvex contact set surrounds the foot on every tilt and blocks ray-escape, so the trichotomy of point types [4, Thm. 2.1(4)] forces drift.

**The Gauss obstruction and the detection frontier.** Reading the ambient certificate $K \neq \operatorname{hconv}(K)$ as an *intrinsic* convexity of $K$ in $\partial\mathcal{H}$ — the form Theorem 6 takes — requires the horosphere geometry, governed by the Gauss equation $K_S = K_M + \det \mathrm{II}$ [4, Prop. 6.1]. This is where the branch develops its own hierarchy, all of it on the *ideal-centre* side of the trichotomy:

- *Flat horospheres* ($K_S \equiv 0$: $\mathbb{H}^n$, and the warped products $M_f$ of variable pinched curvature). Motzkin's theorem applies on the flat horosphere and detection is exactly non-convexity of $K$ — the first reconstruction theorems beyond constant curvature [4, Thm. 8.4].
- CAT(0) *horospheres* (warped over a Hadamard fibre). The sharpness half (convex ⇒ Chebyshev) is unconditional; detection reduces to the Motzkin property of the fibre [4, Thm. 9.3].
- *Positively curved horospheres* (the Heisenberg horosphere of $\mathbb{CH}^2$). Here $K_S$ takes both signs; localisation is carried out exactly in the Carnot metric and detection is *graded non-Chebyshev* — the first reconstruction on a non-CAT(0) horosphere [4, Thm. 10.4].
- *The frontier* ($\mathbb{CH}^n$, $n \geq 3$; pinched sec $\leq -a^2$). Detection is a Chebyshev-set problem in a Carnot group, or Motzkin's theorem in an Alexandrov space of indefinite curvature — open [4, §11].

The entire frontier is a sec $\leq 0$ phenomenon: it lives on the ideal-centre branch, where "defective" is a convexity and the open question is *which* convexity. As §6 will show, it has no counterpart above zero.

## 5 The knife-edge $\sec \equiv 0$: Białożyt's theorem

The flat case is the degenerate boundary of Theorem 3.1, where both descriptions of the centre coincide at infinity (Remark 3.5). It is worth recording exactly what each branch's mechanism becomes there, because it shows Białożyt's theorem (1) as the simultaneous limit of both.

In $\mathbb{R}^n$ the Busemann function $b_u(x) = -\langle x, u\rangle$ is affine (Theorem 3.1(ii)), the maximal free convex region is a half-space $H$, and:

- *From the ideal-centre side:* hconv = conv (the tilts $L_u(y) = \langle y, u^{\perp}\rangle$ are linear, Definition 4.1), so the defect $K \neq \mathrm{hconv}(K)$ is $K \neq \mathrm{conv}(K)$ — Motzkin's non-convexity [12]. The reconstruction is strict: a point on the line through two isolated contacts, beyond their segment, sends its nearest-point ray to infinity with a single foot and is never reconstructed.
- *From the finite-centre side:* the circumcentre of a free ball through two contacts sits at the focal point, which in $\mathbb{R}^n$ has receded to infinity (Remark 3.5); a centre at finite height over a *convex* contact set has a unique foot (convex ⇒ Chebyshev), so it is not medial — which is the same Motzkin condition, seen as the failure of Proposition 3.2 when the equidistant centre is ideal.

**Theorem 5.1** (Białożyt; [1, Thm. 6])**.** *For closed $X \subseteq \mathbb{R}^n$, (1) holds: $\mathcal{R}_X$ is the convex hull less $X$, together with the interiors of the supporting half-spaces whose contact set is non-convex.*

This is the unique curvature on which the certificate is at once *exact*, *strict*, and phrased by ordinary conv. Move down to $\sec < 0$ and conv must become hconv to stay faithful (§4); move up to $\sec > 0$ and the certificate, still formally available, stops being exact and is overwhelmed by total reconstruction (§6). The knife-edge is the only place the nineteenth-century statement is the whole truth.

## 6 The branch $\sec > 0$: finite centres and total reconstruction

Above zero, Theorem 3.1(iii) makes the maximal free convex region a bounded ball with a finite equidistant centre, and Proposition 3.2 collapses detection to a contact count. On the round sphere this collapse is total.

**Theorem 6.1** (Total reconstruction on $\mathbb{S}^n$; [5, Thm. 4.1])**.** *For every closed $X \subseteq \mathbb{S}^n$ with $|X| \geq 2$,*

$$\mathcal{R}_X = \mathbb{S}^n \setminus X.$$

The mechanism is the finite centre throughout. The sharpest case is the *focal pole*: a free hemisphere $H_v^\circ$ whose equator meets $X$ in $\geq 2$ points has, at its pole $v$, a finite centre equidistant at $\frac{\pi}{2}$ from the entire equator, so $v \in M_X$ and $H_v^\circ \subseteq \mathcal{R}_X$ *regardless of the contact set's shape* (Proposition 3.2 with $\rho = \frac{\pi}{2}$) — there is no Motzkin step, no Chebyshev problem, the equatorial geometry idle. Globally, the proof follows the nearest-point ray of any $p \notin M_X$ from its foot $x_0$: it cannot escape (Theorem 3.1(iii): no rays, diameter $\pi$), so before the antipode $x_0$ is overtaken by a rival contact and a finite medial centre appears that still contains $p$. The generalised-gradient flow of $d_X$ [10, 11] makes this unconditional: on the compact $\mathbb{S}^n$ it has finite length, rests only at critical points, and a critical point of reach $< \pi$ carries two distinct nearest points [5, Thm. 4.1, Step 3].

**Proposition 6.2** (The certificate collapses; [5, Prop. 5.1–5.2])**.** *The naive template* (4) *with "defective = non-convex contact set" strictly undercounts on* $\mathbb{S}^n$*. A closed spherical cap* $X = \bar{B}(e, \beta)$*,* $\beta < \frac{\pi}{2}$*, is convex with smooth strictly convex boundary, so no supporting hemisphere is defective and the template predicts* $\varnothing$ *— yet* $\mathcal{R}_X = \mathbb{S}^n \setminus X$*, the whole complement reconstructed from the single focal centre* $-e$*. A latitude circle likewise has every supporting hemisphere touching once, no defective facet, yet its far hemisphere is reconstructed by the focal centre* $-e$*, invisible to the template.*

This is the exact inverse of §4. There the ideal centre sat at infinity and the contact-set defect organised a *strict* $\mathcal{R}_X$; here the centre is an ordinary interior point with no boundary-contact structure to be "defective," it contributes most of $\mathcal{R}_X$, and reconstruction is *total*. The defect notion that runs the negative branch has no positive-curvature counterpart — there is nothing to detect, because everything off $X$ is reconstructed (Remark 3.4 located the sign-flip that causes this).

**Off constant curvature: what survives.** Total reconstruction uses constant curvature twice — the cut point realises the diameter, in every direction at once (Remark 3.5's rigidity). Relaxing it, the conclusion persists under a spread hypothesis and the obstruction is exactly identified.

**Theorem 6.3** (Spread suffices on any compact $M$; [5, Thm. 6.1])**.** *Let* $M$ *be compact and* $X \subseteq M$ *closed and* $\delta$*-dense with* $2\delta < \mathrm{inj}(M)$*. Then* $\mathcal{R}_X = M \setminus X$*. Under* sec $\leq \Delta$ *the hypothesis is explicit via Klingenberg's estimate. For sparse* $X$ *in variable positive curvature the sole obstruction is* cut-shielding *— a ray reaching the cut locus of its foot before any rival contact — and total reconstruction holds iff cut-shielding is absent [5, Prop. 7.1]; whether it can occur for sparse* $X$ *on a positively curved* $M$ *is open.*

**Theorem 6.4** (The two-geodesic axis removes the obstruction; [5, Thm. 8.4])**.** *Let* $M_X^g$ *be the set of points with two distinct minimising geodesics to* $X$ *(the cut locus of* $X$*) and* $\mathcal{R}_X^g = \bigcup_{a \in M_X^g} B(a, r(a))$*. Then for* every *compact* $M$ *and closed* $X$ *with* $|X| \geq 2$*,* $\mathcal{R}_X^g = M \setminus X$*, with no spread, curvature, or genericity hypothesis; and* $M_X^g = M_X$ *on* $\mathbb{R}^n$ *and* $\mathbb{S}^n$*, so the sphere and Euclidean theories are unchanged. The gradient flow always rests at a critical point, which carries two minimising geodesics by definition; cut-shielding is precisely the case where these have a common endpoint, and the two-geodesic axis counts it as medial.*

Both refinements stay on the finite-centre side: the flow lands at a *finite* critical point, and the only question is whether its two geodesics reach two *points* (then $M_X$, Theorem 6.3) or one (then only $M_X^g$, Theorem 6.4). The convexity detection of the negative branch never reappears, exactly as the trichotomy predicts.

## 7 The unified picture

**The phenomenon, in one line.** A point escapes reconstruction by sending its nearest-point ray to infinity with a single foot. Whether this can happen is whether a free convex region can be unbounded, which is the sign of the Busemann Hessian, which is the sign of the curvature. Below zero it happens, $\mathcal{R}_X$ is partial and organised by a convexity defect of contact sets; above zero compactness forbids it, every point is swallowed by a finite centre, $\mathcal{R}_X = M \setminus X$; at zero the ray escapes but only just, and Białożyt's conv-certificate is exact.

**The phase transition.** The same template (4) carries all three regimes, with one parameter — the centre type — flipping at sec $= 0$:

| sec | Hess $b$ | region $\mathcal{H}$ | centre | detection / $\mathcal{R}_X$ |
|---|---|---|---|---|
| $\le 0$ | $\succeq 0$ | convex, unbounded | ideal $\xi \in \partial_\infty M$ | non-Chebyshev hull; $\mathcal{R}_X \subsetneq M \setminus X$ |
| $\equiv 0$ | $\equiv 0$ | half-space | $\infty$ | Motzkin (non-convex); strict |
| $> 0$ | $\preceq 0$ | bounded ball | finite $a \in M$ | $\ge 2$ contacts; $\mathcal{R}_X = M \setminus X$ (total) |

The left two columns are Theorem 3.1; the right two are Propositions 3.3 and 3.2 with Theorems 4.2, 5.1, 6.1. Reading down the "detection" column is the content of Theorem 6's generalisation: it is *not* one formula but a convexity question that trivialises into a counting question as the ideal centre descends to a finite one.

**What each branch leaves open.** The two open problems are the two branches' own, and the trichotomy says they do not mix.

- *Negative branch (ideal centre).* Detection on non-CAT(0) horospheres: a Chebyshev-set problem in a Carnot group ($\mathbb{CH}^n$), or Motzkin's theorem in an Alexandrov space of indefinite curvature (pinched sec $\le -a^2$) [4, §11]. This is a *convexity* question and exists only because the centre is ideal.

- *Positive branch (finite centre).* Non-detection for sparse $X$ in *variable* positive curvature: whether cut-shielding can obstruct total reconstruction [5, §7]. This is a *cut-locus* question and exists only because the certificate has collapsed — there is no convexity content left, only the geometry of the cut locus. The two-geodesic axis removes it entirely (Theorem 6.4).

The trichotomy thus also explains why the search for a positive-curvature mirror of the $\mathbb{CH}^2$ detection theorem fails: detection is an ideal-centre structure, and above zero there are no ideal centres [5, Thm. 3.4].

**The natural sequel.** Total reconstruction makes the *set* problem trivial above zero, which points to the *domain* reconstruction problem of Lieutier [10]: reconstruct an open $\Omega \subseteq \mathbb{S}^n$ from the medial axis of $\partial\Omega$ using only balls contained in $\Omega$. The confinement excludes the far-side focal centres, the antipodal collapse does not occur, and a genuine convexity/defect theory — now with the positively curved induced geometry of $\partial\Omega$, an Alexandrov rather than horosphere setting — should re-emerge. That is where the two branches of this note would finally meet on the same side of zero.

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